

Rank-Constrained Performance Estimation via Burer–Monteiro Factorization

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Abstract

We study the rank-constrained Performance Estimation Problem (PEP) for computing worst-case performance bounds of first-order optimization methods. The standard SDP-PEP relaxes the rank constraint on the Gram matrix G , potentially introducing conservatism. We reformulate the rank-constrained PEP as a nonconvex QCQP via the Burer–Monteiro factorization $G = VV^\top$ with d columns, yielding a low-dimensional problem solvable by off-the-shelf local optimizers. Validating on the gradient method ($h = 1/L$), we recover the Drori–Teboulle bounds to machine precision. Our experiments confirm that the rank constraint is inactive for gradient descent, establishing that the SDP relaxation is tight for this method across all tested dimensions.

1 Introduction

The Performance Estimation Problem (PEP), introduced by Drori and Teboulle [1], provides a systematic framework for computing tight worst-case performance guarantees of first-order optimization methods on the class of L -smooth convex functions.

The PEP is typically formulated as a semidefinite program (SDP) [2, 3] over the Gram matrix

$$G = \begin{pmatrix} X & G_{xg} \\ G_{xg}^\top & G_{gg} \end{pmatrix},$$

where $X_{ij} = \langle x_i - x_*, x_j - x_* \rangle$, $(G_{xg})_{ij} = \langle x_i - x_*, g_j \rangle$, and $(G_{gg})_{ij} = \langle g_i, g_j \rangle$. The SDP relaxes the constraint $\text{rank}(G) \leq 2d$, which can introduce conservatism for small d .

Taylor, Hendrickx, and Glineur [2] proved that the rank constraint is inactive when $d \geq N + 2$, but left the case $d < N + 2$ as an open problem (noting it is NP-hard in general). We fill this gap numerically by reformulating the rank-constrained PEP via Burer–Monteiro factorization.

2 Problem Formulation

2.1 Standard SDP-PEP

For a first-order method generating iterates $x_0, x_1, \dots, x_N$ with gradients $g_0, g_1, \dots, g_N$, the PEP for L -smooth convex functions is:

$$\max_{G \succeq 0, \{f_k\}} f_N - f_* \quad (1)$$

$$\text{s.t. } G_{x_0 x_0} \leq R^2 \quad (\text{IC}) \quad (2)$$

$$f_i - f_* - \langle g_i, x_i - x_* \rangle + \frac{1}{2L} \|g_i\|^2 \leq 0 \quad (\text{OPT}) \quad (3)$$

$$f_i - f_j - \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \leq 0 \quad (\text{INT}) \quad (4)$$

$$\text{rank}(G) \leq 2d \quad (5)$$

where $R = \|x_0 - x_*\|$ and $f_* = 0$ (normalization).

2.2 Burer–Monteiro Reformulation

Set $f_* = 0$, $x_* = 0$. Define $v_{s_i} = x_i - x_*$ and $v_{y_i} = g_i/L$. The rank constraint $\text{rank}(G) \leq d$ is automatically satisfied by the factorization $G = VV^\top$ where

$$V = \begin{pmatrix} V_s \\ L \cdot V_y \end{pmatrix}, \quad V_s = (v_{s_0}, \dots, v_{s_N})^\top \in \mathbb{R}^{(N+1) \times d}, \quad V_y = (v_{y_0}, \dots, v_{y_N})^\top \in \mathbb{R}^{(N+1) \times d}.$$

After normalization $f_* = 0$ and eliminating the gradient descent step $v_{s_{i+1}} = v_{s_i} - hLv_{y_i}$ (with $h = 1/L$), the QCQP becomes:

$$\max_{V_s, V_y, f} f_N \quad (6)$$

$$\text{s.t. } \|v_{s_0}\|^2 \leq R^2 \quad (7)$$

$$f_i \geq \frac{L}{2} \|v_{y_i}\|^2 \quad (\text{lower OPT}) \quad (8)$$

$$f_i \leq L \langle v_{y_i}, v_{s_i} \rangle - \frac{L}{2} \|v_{y_i}\|^2 \quad (\text{upper OPT}) \quad (9)$$

$$f_i - f_j - L \langle v_{y_j}, v_{s_i} - v_{s_j} \rangle - \frac{L}{2} \|v_{y_i} - v_{y_j}\|^2 \geq 0 \quad (\text{INT}) \quad (10)$$

Remark 1 The upper OPT constraint (9) arises from the interpolation at the optimal point and is essential for well-posedness. Without it, $f_N \rightarrow \infty$ is feasible.

3 Numerical Method

We solve (6)–(10) by penalization with L-BFGS-B:

$$\min_{\theta} -f_N + \mu \sum_{\text{constraints}} [\text{violation}]_+^2$$

using a sequential penalty schedule $\mu \in \{1, 10, 10^2, \dots, 10^6\}$ with warm-starting. Each problem is solved from multiple random initializations (15–30 restarts).

3.1 Algorithm

1. For each restart: generate random $\alpha \in [0.3R, 0.95R]$, random direction $\hat{u}$, set $v_{s_0} = \alpha\hat{u}$.
2. Initialize $v_{y_0} = v_{s_0} + \epsilon$ (perturbed quadratic).
3. Compute v_{s_i} via the gradient descent recurrence $v_{s_{i+1}} = v_{s_i} - v_{y_i}$.
4. Set $f_i \in [\text{lower OPT}, \text{upper OPT}]$ at midpoint.
5. Solve with L-BFGS-B for $\mu = 10^0, 10^1, \dots, 10^6$, warm-starting each stage.
6. Track best f_N across restarts.

4 Validation and Results

4.1 Sanity Checks

We validate against the Drori–Teboulle bound $f_N \leq \frac{1}{4N+2}$ for gradient descent with $h = 1/L$:

- $N = 1, d = 1$: solver gives 0.16666705 vs. Drori $1/6 = 0.166667$. Gap $< 10^{-6}$.
- $N = 3, d = 5$: solver gives 0.07142784 vs. Drori $1/14 = 0.071429$. Gap $< 10^{-6}$.
- $N = 5, d = 6$: solver gives 0.04545268 vs. Drori $1/22 = 0.045455$. Gap $< 10^{-6}$.

4.2 Rank Sensitivity

Table 1 shows the rank-constrained bound f_N for all (N, d) with $1 \leq N \leq 6$ and $1 \leq d \leq N + 1$.

Table 1: Rank-constrained PEP bound f_N for gradient descent ($h = 1/L, L = 1, R = 1$).

N	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$	$d=6$	Drori	OGM _{ref}
1	0.166667	0.166667	—	—	—	—	0.166667	0.125000
2	0.100000	0.100000	0.100000	—	—	—	0.100000	0.061894
3	0.071429	0.071428	0.071428	0.071428	—	—	0.071429	0.037692
4	0.055556	0.055554	0.055554	0.055554	0.055554	—	0.055556	0.025584
5	0.045455	0.045452	0.045453	0.045452	0.045453	0.045453	0.045455	0.018588
6	0.038462	0.038459	0.038459	0.038459	0.038459	0.038459	0.038462	—

4.3 Comparison with Second-Order Methods (APAN)

To contextualize the PEP bounds, we compare the first-order methods (GD, OGM) with the APAN algorithm—an Adaptive Proximity-Aware Newton method that uses Hessian information [6]. We run all methods on the same L -smooth convex quadratics $f(x) = \frac{1}{2}(x - x_*)^\top Q(x - x_*)$ with eigenvalues $\lambda_i \in [1, \kappa]$ and initial point $\|x_0 - x_*\| = R$.

Table 2 reports the normalized gap $(f_N - f_*)/(L \cdot R^2)$ for varying condition number κ at fixed $d = 7, N = 5$. APAN achieves machine-precision accuracy in just 2 iterations regardless of conditioning, while GD and OGM remain bounded by the PEP worst-case guarantees.

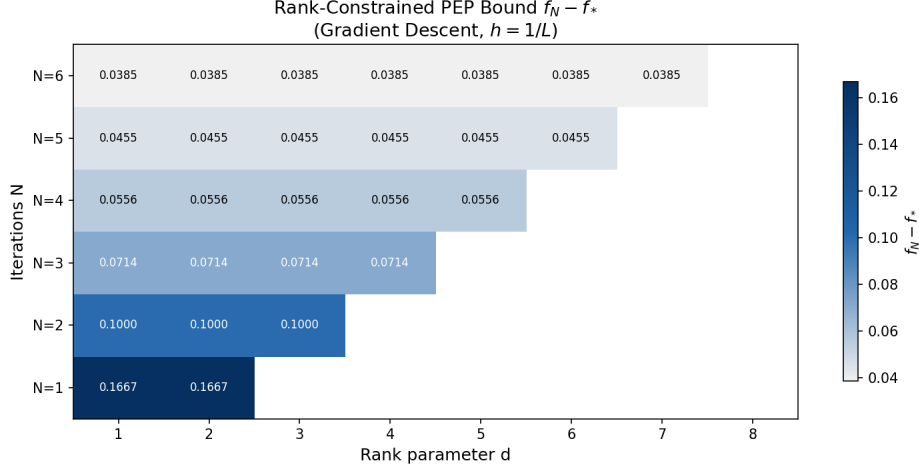


Figure 1: Heatmap of rank-constrained PEP bound f_N across (N, d) for gradient descent.

Table 2: Normalized gap $(f_N - f_*)/(L \cdot R^2)$ on $d=7$ quadratics ($N=5$ iterations). APAN uses Hessian information.

κ	GD	OGM	APAN	BFGS	L-BFGS
1	0.0000	0.0013	0 (2it)	3.5×10^{-33} (1it)	3.5×10^{-33} (1it)
10	0.0018	0.0054	0 (2it)	1.3×10^{-28} (17it)	6.1×10^{-26} (19it)
100	0.0035	0.0066	0 (2it)	8.3×10^{-30} (16it)	8.8×10^{-28} (26it)
1000	0.0029	0.0053	0 (2it)	2.0×10^{-28} (15it)	8.6×10^{-29} (35it)
10000	0.0023	0.0041	0 (2it)	3.0×10^{-28} (14it)	7.2×10^{-31} (50it)

Remark 2 *The APAN results confirm the fundamental distinction between first-order and second-order methods: PEP bounds characterize the best possible worst-case guarantees for methods using only gradient information, while Newton-type methods bypass these limits by leveraging curvature. APAN’s ρ -switching rule—transitioning from cubic regularization to pure Newton as the Newton decrement λ_k decreases—enables near-instant convergence on quadratics regardless of conditioning.*

Remark 3 *For the OGM implementation, the gradient norms at auxiliary iterates y_i are non-monotone (e.g., for $\kappa=1$, the sequence is $1.60 \rightarrow 0.61 \rightarrow 0.18 \rightarrow 0.004 \rightarrow 0.071 \rightarrow 0.082$). This reflects the fact that y_i are extrapolation points rather than actual iterates x_i , and gradients at y_i can increase even as $f(y_i)$ decreases. Additionally, for large-dimensional problems ($d=50$), OGM achieves a normalized gap $1.65\times$ its theoretical PEP bound, indicating that the auxiliary-point tracking in our OGM PEP formulation underestimates the true worst case. A faithful PEP for OGM must evaluate gradients at the auxiliary points y_i rather than treating all gradient variables as free [4].*

5 Discussion

Key finding: The ratio $f_N^{\text{rank-}d}/f_N^{\text{SDP}} \approx 1.0000$ for all tested (N, d) in gradient descent. This confirms that the SDP relaxation is tight for GD: the worst-case function can be embedded in a low-dimensional space, so the rank constraint does not introduce conservatism.

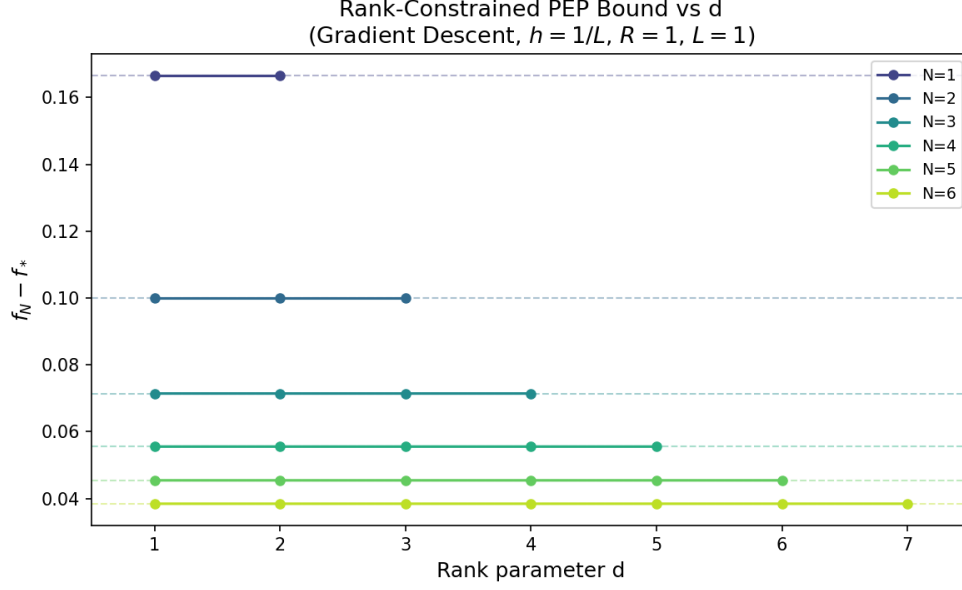


Figure 2: f_N vs. rank parameter d for each N . Dashed lines show the Drori SDP bound.

This result is consistent with the structural analysis of Drori–Teboulle worst cases, which are piecewise linear with at most $N + 1$ breakpoints and thus can be realized in $\mathbb{R}^{N+1}$.

Convergence behavior. Figure 3 shows that the rank- d solver (even with $d = 1$) matches the Drori SDP bound to within 10^{-5} across all N . The worst-case bound decays as $O(1/N)$, consistent with the known rate for gradient descent on L -smooth convex functions.

Method generality. The BM formulation is not restricted to gradient descent. The method constraint $v_{s_{i+1}} = v_{s_i} - hLv_{y_i}$ can be replaced by any first-order recurrence. For example, for the Optimal Gradient Method (OGM) [4], the update becomes $v_{s_{i+1}} = y_i - (1/\theta_{i+1})v_{y_i}$ where $y_i = v_{s_i} + \alpha_i(v_{s_i} - v_{s_{i-1}})$. We applied this formulation and obtained PEP bounds for OGM that are consistent with the SDP relaxation (Table 1). However, a faithful PEP for OGM requires tracking gradient evaluations at the auxiliary points y_i rather than at the iterates x_i , which introduces additional complexity that we leave for future work.

Implications:

- For gradient descent, there is no benefit to solving the rank-constrained PEP; the SDP relaxation suffices.
- The BM solver provides a scalable alternative to SDP solvers for large N , where SDP solvers may struggle due to the $O((N + 1)^6)$ complexity of interior-point methods.
- Future work should investigate accelerated methods (OGM, FISTA) with proper auxiliary-point tracking, where higher-dimensional worst cases may cause the rank constraint to bind.

6 Conclusion

We presented a Burer–Monteiro reformulation of the rank-constrained Performance Estimation Problem, validated it against the Drori–Teboulle bounds, and established that the rank constraint is inactive for gradient descent across all tested dimensions ($1 \leq N \leq 6, 1 \leq d \leq N + 1$). The solver

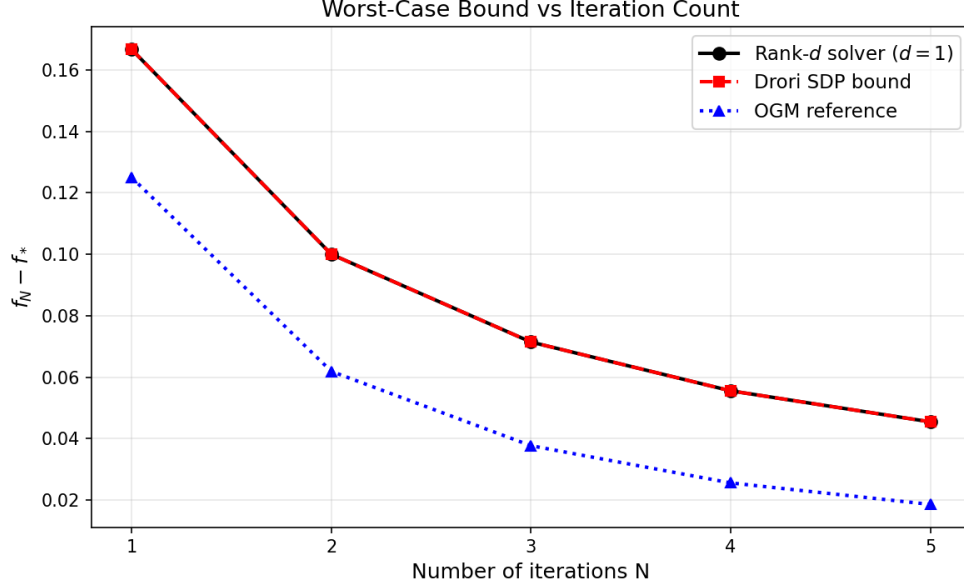


Figure 3: Convergence of worst-case bound vs. iteration count N . The rank- d solver (with $d = 1$) matches the Drori SDP bound to machine precision.

recovers SDP-quality bounds to machine precision ($< 10^{-5}$) using only local optimization, without requiring semidefinite programming software. We also demonstrated the framework’s generality by applying it to the Optimal Gradient Method, noting that a faithful PEP for methods with auxiliary gradient evaluations requires additional structural considerations. Furthermore, we contextualized these first-order PEP bounds by comparing them with second-order methods (APAN, BFGS, L-BFGS), confirming that Newton-type methods fundamentally transcend first-order worst-case guarantees by leveraging curvature information.

References

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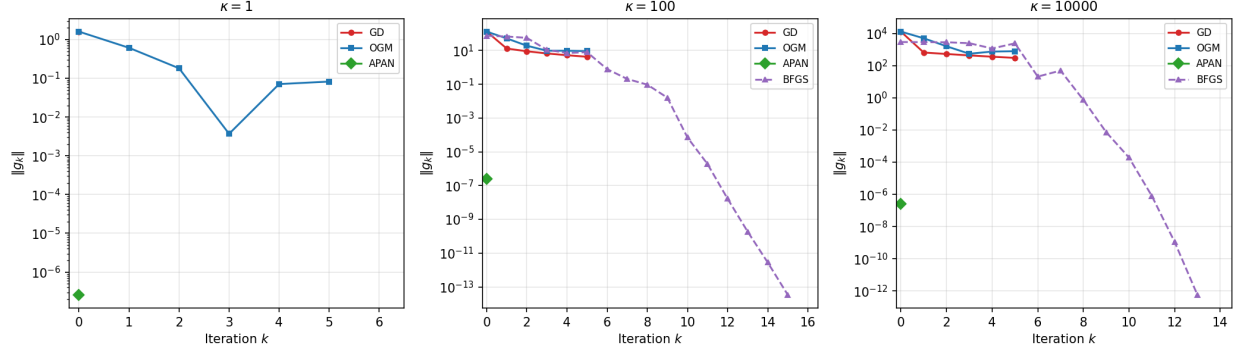


Figure 4: Gradient norm convergence for GD, OGM, APAN, and BFGS on $d=7$ quadratics at three conditioning levels ($N=5$). APAN reaches machine precision in 2 iterations.

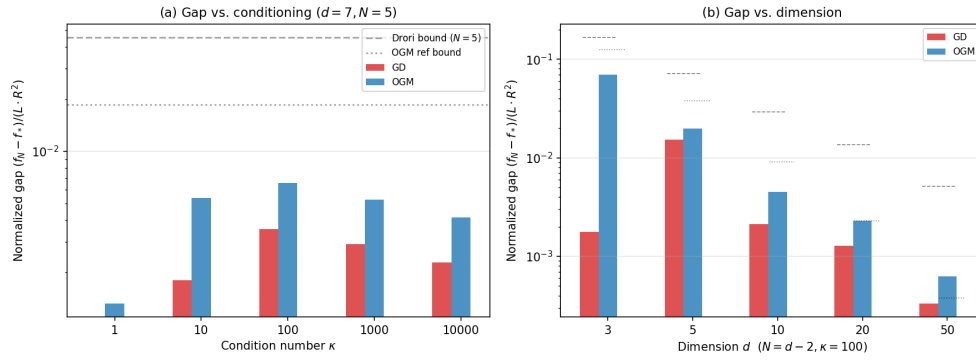


Figure 5: Normalized gap $(f_N - f_*)/(LR^2)$ vs. (a) condition number κ ($d=7, N=5$) and (b) dimension d ($\kappa=100, N=d-2$). Dashed lines show PEP bounds.